

Fundamentals of Deep Learning and Programming

Lecture 0: Course Overview

Fall Semester 2025

Adjunct Professor: Donghoon Kang

Course Overview

September

Week 1

Course overview

Introduction to Python Programming

Week 2

Neural Network Basics

Week 3

Training Neural Networks

PyTorch

Week 4

Convolutional Neural Networks

Course Overview

October

Week 5

Object Detection

theory

Week 6

Object Detection

coding

Week 7

Introduction to NLP (Natural Language Processing)

Week 8

Attention Mechanism

Week 9

Mid term exam

Course Overview

November

Week 10

Transformer Architecture

theory

Week 11

Transformer Architecture

coding

Week 12

Generative Models – Introduction to GANs, VAEs

Week 13

Diffusion Models

theory

Part 1

Course Overview

December

Week 14

Diffusion Models

theory

Part 2

Week 15

Diffusion Models

coding

Week 16

Final exam